PROBLEM STATEMENT:

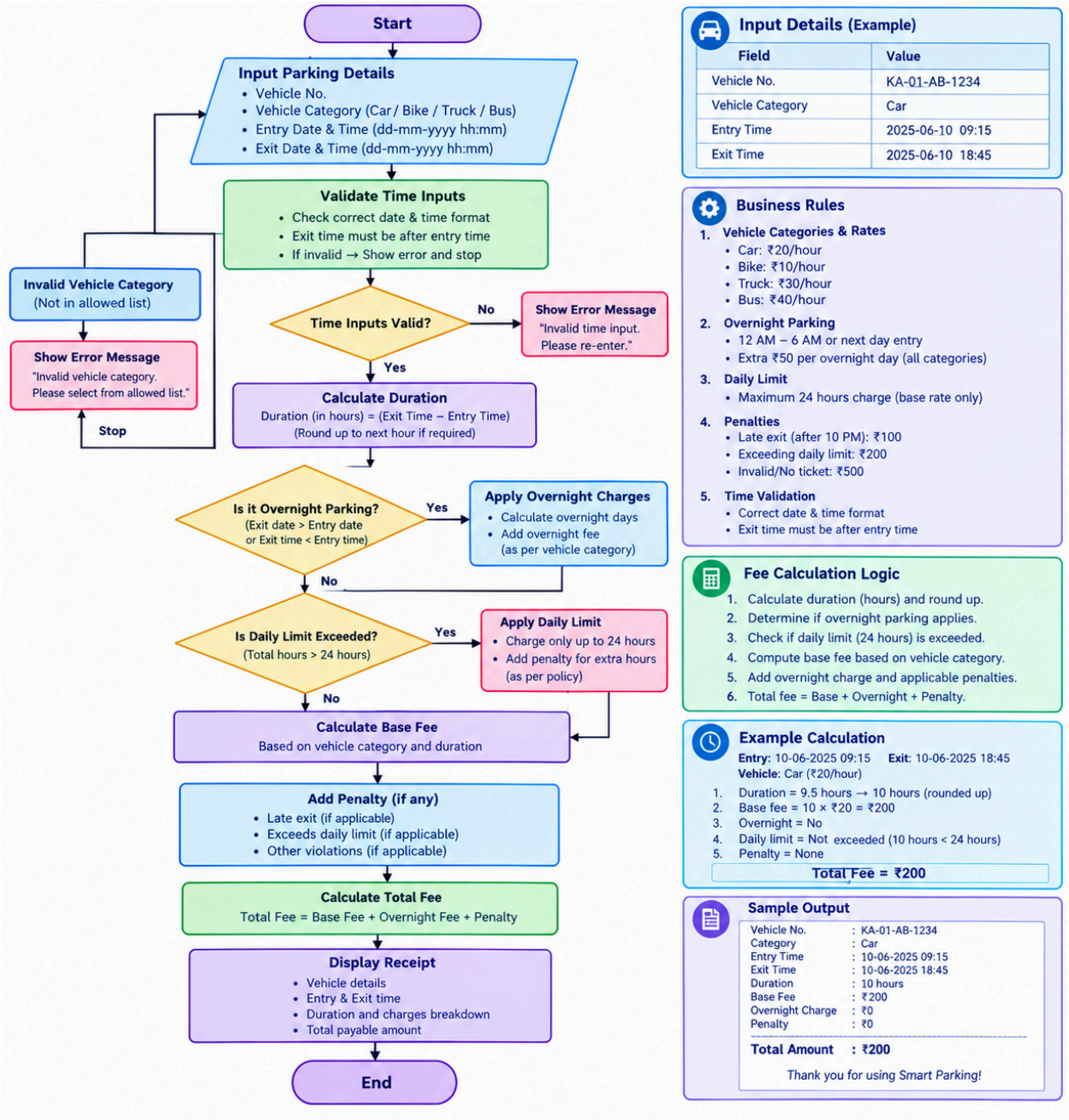
OBJECTIVE:

To develop a program that calculates parking fees based on vehicle category, entry time, exit time, overnight parking, daily fee limits, and penalty conditions. The program validates time inputs, calculates the total parking duration, applies the appropriate charges, and displays the final parking bill.

ALGORITHM:

1. Start the program.
2. Read the vehicle category (Car, Bike, or Truck).
3. Read the entry date and time and exit date and time.
4. Validate the date and time inputs.
5. If the exit time is earlier than the entry time, display an invalid time message and stop.
6. Calculate the total parking duration in hours.
7. Determine whether overnight parking has occurred.
8. Calculate the parking fee according to the vehicle category and hourly rate.
9. Apply the daily maximum fee limit for each parking day.
10. Add any applicable penalty charges.
11. Calculate the final fee.
12. Display the vehicle category, parking duration, overnight status, parking fee, penalty, and total amount.
13. Stop.

FLOWCHART:



PROGRAM:

Assumed fee policy: Bike ₹20/hour (₹100 daily limit), Car ₹50/hour (₹300 daily limit), Truck ₹80/hour (₹500 daily limit). An overnight surcharge of ₹100 applies if parking crosses midnight,

and a penalty of ₹200 applies when parking exceeds 24 hours. Each started hour is charged as a full hour.

```
#include <stdio.h>
#include <time.h>
#include <string.h>
#include <math.h>

int main() {
    char vehicle[20], entryDate[11], exitDate[11];
    char entryTime[6], exitTime[6];
    int rate, dailyLimit, overnight = 0, penalty = 0;
    int eh, em, xh, xm;
    int ey, en, ed, xy, xn, xd;
    int days, hours, totalMinutes, billableHours;
    int dailyFee, parkingFee, totalFee;
    struct tm entry = {0}, exit = {0};
    time_t t1, t2;

    printf("Vehicle category (Bike/Car/Truck): ");
    scanf("%19s", vehicle);

    if (strcmp(vehicle, "Bike") == 0) {
        rate = 20;
        dailyLimit = 100;
    } else if (strcmp(vehicle, "Car") == 0) {
        rate = 50;
        dailyLimit = 300;
    } else if (strcmp(vehicle, "Truck") == 0) {
        rate = 80;
        dailyLimit = 500;
    } else {
        printf("Invalid vehicle category!\n");
        return 0;
    }

    printf("Entry date (YYYY-MM-DD): ");
    scanf("%10s", entryDate);
    printf("Entry time (HH:MM): ");
    scanf("%5s", entryTime);

    printf("Exit date (YYYY-MM-DD): ");
    scanf("%10s", exitDate);
```

```
printf("Exit time (HH:MM): ");
scanf("%5s", exitTime);

if (sscanf(entryDate, "%d-%d-%d",
          &ey, &en, &ed) != 3 ||
    sscanf(exitDate, "%d-%d-%d",
          &xy, &xn, &xd) != 3 ||
    sscanf(entryTime, "%d:%d", &eh, &em) != 2 ||
    sscanf(exitTime, "%d:%d", &xh, &xm) != 2) {
    printf("Invalid date or time format!\n");
    return 0;
}

if (ey < 1900 || en < 1 || en > 12 ||
    ed < 1 || ed > 31 ||
    xy < 1900 || xn < 1 || xn > 12 ||
    xd < 1 || xd > 31 ||
    eh < 0 || eh > 23 || em < 0 || em > 59 ||
    xh < 0 || xh > 23 || xm < 0 || xm > 59) {
    printf("Invalid date or time values!\n");
    return 0;
}

entry.tm_year = ey - 1900;
entry.tm_mon = en - 1;
entry.tm_mday = ed;
entry.tm_hour = eh;
entry.tm_min = em;
entry.tm_isdst = -1;

exit.tm_year = xy - 1900;
exit.tm_mon = xn - 1;
exit.tm_mday = xd;
exit.tm_hour = xh;
exit.tm_min = xm;
exit.tm_isdst = -1;

t1 = mktime(&entry);
t2 = mktime(&exit);

if (t1 == (time_t)-1 || t2 == (time_t)-1) {
    printf("Invalid date!\n");
    return 0;
}
```

```
/* Check normalized dates to reject impossible dates */
if (entry.tm_year != ey - 1900 ||
    entry.tm_mon != en - 1 ||
    entry.tm_mday != ed ||
    exit.tm_year != xy - 1900 ||
    exit.tm_mon != xn - 1 ||
    exit.tm_mday != xd) {
    printf("Invalid calendar date!\n");
    return 0;
}

if (t2 <= t1) {
    printf("Invalid exit time! Exit must be after entry.\n");
    return 0;
}

totalMinutes = (int)difftime(t2, t1) / 60;
hours = totalMinutes / 60;
billableHours = (totalMinutes + 59) / 60;

days = (int)((t2 - t1 + 86399) / 86400);
if (days < 1)
    days = 1;

overnight = (xy != ey || xn != en || xd != ed);

parkingFee = billableHours * rate;
dailyFee = dailyLimit * days;

if (parkingFee > dailyFee)
    parkingFee = dailyFee;

if (totalMinutes > 24 * 60)
    penalty = 200;

totalFee = parkingFee + penalty;

if (overnight)
    totalFee += 100;

printf("\n----- PARKING BILL -----\n");
printf("Vehicle Category : %s\n", vehicle);
printf("Parking Duration : %d hours %d minutes\n",
```

```
        hours, totalMinutes % 60);
    printf("Overnight Parking: %s\n",
        overnight ? "Yes" : "No");
    printf("Parking Fee      : Rs.%d\n", parkingFee);
    printf("Overnight Charge : Rs.%d\n",
        overnight ? 100 : 0);
    printf("Penalty          : Rs.%d\n", penalty);
    printf("Total Fee        : Rs.%d\n", totalFee);

    return 0;
}
```

PSEUDOCODE:

```
BEGIN

INPUT vehicle category

IF vehicle is Bike
   rate = 20
   dailyLimit = 100
ELSE IF vehicle is Car
   rate = 50
   dailyLimit = 300
ELSE IF vehicle is Truck
   rate = 80
   dailyLimit = 500
ELSE
   PRINT "Invalid vehicle category"
   STOP
END IF

INPUT entry date and time
INPUT exit date and time

VALIDATE date and time formats and values

IF date or time is invalid
   PRINT "Invalid date or time"
   STOP
END IF

CONVERT entry and exit to timestamps
```

```
IF exit timestamp <= entry timestamp
   PRINT "Invalid exit time"
   STOP
END IF

CALCULATE total parking minutes
billableHours = CEILING(totalMinutes / 60)
days = CEILING(totalMinutes / 1440)

parkingFee = billableHours * rate
maximumFee = days * dailyLimit

IF parkingFee > maximumFee
   parkingFee = maximumFee
END IF

IF entry date differs from exit date
   overnightCharge = 100
ELSE
   overnightCharge = 0
END IF

IF totalMinutes > 1440
   penalty = 200
ELSE
   penalty = 0
END IF

totalFee = parkingFee + overnightCharge + penalty

DISPLAY parking duration
DISPLAY parking fee
DISPLAY overnight charge
DISPLAY penalty
DISPLAY total fee

END
```

OUTPUT:

Example 1 — Car parked overnight

Vehicle category (Bike/Car/Truck): Car
Entry date (YYYY-MM-DD): 2026-10-26
Entry time (HH:MM): 22:00
Exit date (YYYY-MM-DD): 2026-10-27
Exit time (HH:MM): 02:30

----- PARKING BILL -----
Vehicle Category : Car
Parking Duration : 4 hours 30 minutes
Overnight Parking: Yes
Parking Fee : Rs.250
Overnight Charge : Rs.100
Penalty : Rs.0
Total Fee : Rs.350

Example 2 — Invalid exit time

Vehicle category (Bike/Car/Truck): Bike
Entry date (YYYY-MM-DD): 2026-10-26
Entry time (HH:MM): 10:00
Exit date (YYYY-MM-DD): 2026-10-26
Exit time (HH:MM): 09:00